

Chathurya V R 251801045-2025-AIDS ▾C2**Started on** Thursday, 17 September 2026, 8:25 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:26 AM**Time taken** 32 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5	1
1 1 2 3 4	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  #define MAX_N 100005
4
5  int arr[MAX_N];
6
7  int main() {
8      int n;
9
10     // Read the total number of elements
11     if (scanf("%d", &n) != 1) return 0;
12
13     // Read the n elements sequentially
14     for (int i = 0; i < n; i++) {
15         if (scanf("%d", &arr[i]) != 1) return 0;
16     }
17
18     // Phase 1: Find the intersection point of the tortoise and the hare
19     int tortoise = arr[0];
20     int hare = arr[arr[0]];
21
22     while (tortoise != hare) {
23         tortoise = arr[tortoise];
24         hare = arr[arr[hare]];
25     }
26
27     // Phase 2: Find the entrance to the cycle (the duplicate value)
28     tortoise = 0;
29     while (tortoise != hare) {
30         tortoise = arr[tortoise];
31         hare = arr[hare];
32     }
33
34     // Output the duplicate element matching the expected output format
35     printf("%d\n", tortoise);
36
37     return 0;
38 }
39

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)